

Design document

Group Nr. 3

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Contents

1 Project description

1.1 Problem description

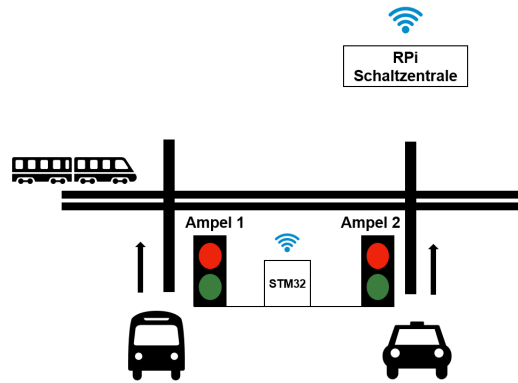


Figure 1: system description

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2 Software architecture and design

2.1 Software modules

Safety related modules

1. Modul x:
Description:
Functions:
Data:
Requirements see: ??,

Security related modules

Modules with no influence on Safety and Security

2.2 Libraries

Description of used function with parameters.

2.3 Interrupts

Definition of priorities.

2.4 Pinout

3 Program flowchart

Hier bitte Sequenzdiagramme, bzw. Programmablaufdiagramme

4 Hazard identification

The only source of physical harm in this system is the rotating actor (rotor). The safety function of the demonstrator is: if an object is closer to the rotor than the speed-dependent minimum distance, the motor must be switched off. The hazards below are therefore all situations in which this safety function is lost, is applied too late, or is based on wrong data.

4.1 Identified hazards and countermeasures

1. Hazard 1: Undetected failure of a distance sensor

Cause: An HC-SR04 delivers no echo at all (broken wiring, defective transducer), returns a permanently frozen value, returns a value outside the specified measuring range, or receives the echo of the second sensor (cross-talk).

Effect: The control unit keeps working with an invalid or outdated distance value and interprets the danger zone as free. The motor is not switched off although a person has already entered the protected area.

Countermeasure in software (Raspberry Pi 2):

- Echo timeout monitoring: if no echo edge is received within the maximum expected time, the measurement is marked as invalid. An absent echo is never interpreted as “no object”.
- Range check: measurements outside the specified measuring range of the sensor are rejected.
- Stuck-value detection: if a sensor reports an unchanged value over a defined number of consecutive cycles, it is treated as faulty.
- Plausibility check between both sensors: if the difference of the two measurements exceeds a defined limit, no trustworthy distance information exists.
- Age check: every distance value carries a timestamp. A value older than one monitoring cycle is discarded and never reused. No distance variable is evaluated before a valid measurement has been written to it.
- Time multiplexing: only one sensor is triggered at a time, so an echo can always be assigned to exactly one sensor.
- Fail-safe reaction: each of the checks above leads to the safe state (motor off). The error state is latched and displayed; the system restarts only after a manual acknowledgement.

Requirement see ??

2. Hazard 2: Protection zone too small for the actual rotor speed

Cause: The minimum distance is derived from the speed setpoint that is entered on the host unit and transmitted via Bluetooth. The actual rotor speed can deviate from this value during run-up, under changing load, after a corrupted or outdated transmission, or because the rotor continues to coast down after the motor has been switched off.

Effect: The monitored area is smaller than required for the real speed, or the system reports the safe state while the rotor is still rotating. A person can reach the rotor while it still carries dangerous energy.

Countermeasure in software (Raspberry Pi 2):

- The minimum distance is calculated from the measured rotor speed, not from the transmitted setpoint. If no valid speed measurement is available, the maximum speed is assumed as the worst case.
- The minimum distance includes the reaction time of the monitoring cycle and the coast-down time until standstill, not only the geometric distance to the rotor.
- Setpoints received via Bluetooth are accepted only with a valid checksum, within the allowed value range and with a current timestamp. An outdated setpoint is discarded.
- A speed increase is only executed after the correspondingly larger protection zone has been verified as free.
- The safe state is signalled only when the measured speed is zero, not when the switch-off command has been issued.

Requirement see ??

3. **Hazard 3: Loss of cyclic monitoring while the motor is running**

Cause: Software fault on the control unit: endless or blocked loop (for example waiting for an echo that never arrives), deadlock, crash of the control process, or an exceeded cycle time caused by system load.

Effect: The motor keeps its last enable state while the protected area is no longer monitored. The safety function is lost silently, without any visible reaction of the system.

Countermeasure in software (Raspberry Pi 2):

- Hardware watchdog that is retriggered only at the end of a completely executed monitoring cycle. An expired watchdog resets the controller; the enable output then defaults to “off”.
- Dynamic enable signal: the motor driver is enabled by a periodic signal that only a running and complete cycle can generate. A static signal level cannot keep the motor switched on.
- Cycle time monitoring against a defined deadline. A missed deadline leads to the safe state.
- All read operations on sensors and on the communication interface use timeouts. There are no unbounded blocking waits in the safety related path.
- Self-test of the complete safety path at program start (sensor reading, enable output, error indication) before the motor can be started for the first time.

Requirement see ??

4.2 Identified hazards without countermeasures

1. Hazard 1: Mechanical failure of the rotor

Cause: Material fatigue, a loose fastening, imbalance at high speed or a foreign object entering the rotor.

Effect: Parts are ejected with high energy and can cause injuries also outside the monitored area.

Reason why no countermeasure is implemented: The event occurs within milliseconds and cannot be detected by distance sensors. Software can neither prevent it nor reduce its consequences. Only mechanical measures such as an enclosure, a defined mounting torque and a limited maximum speed are effective, and these are outside the software scope of this project.

Residual risk: Accepted and documented. The demonstrator is operated with a limited maximum speed.

2. Hazard 2: Failure of the motor driver (switch-off command without effect)

Cause: A welded or short-circuited switching element in the power stage, a defective enable input, or a driver that remains supplied although the control unit commands the switch-off.

Effect: The motor keeps running although the software has correctly detected the hazard and has commanded the shutdown. A single hardware fault bypasses the complete safety function.

Reason why no countermeasure is implemented: The software cannot detect or override a switching element that is stuck in the conducting state. A second, independent shutdown path would be required, for example a safety relay in the supply line with feedback contact monitoring, or a driver with a dedicated safe torque off input. The demonstrator uses a single-channel shutdown path; adding redundant hardware is outside the scope of this project.

Residual risk: Documented single point of failure.

3. Hazard 3: Physical detection limits of the ultrasonic sensors

Cause: The HC-SR04 evaluates reflected sound and covers only a narrow detection cone. Objects approaching from the side, from above or from below are outside this cone. Sound absorbing materials such as soft clothing, gloves or foam, and strongly inclined surfaces that deflect the echo, produce no usable reflection. Small objects such as fingers, an approach faster than the measurement cycle, other ultrasonic sources and strong air movement further reduce the detection reliability.

Effect: A person can enter the danger zone without being detected. The motor is not switched off, although the software works correctly.

Reason why no countermeasure is implemented: These are properties of the measuring principle and not software faults. No plausibility check can detect an object that produces no echo at all. A reliable solution would require a certified safety sensor with a defined detection capability and internal self-monitoring, for example a safety laser scanner or a light curtain. The HC-SR04 is a standard component without any safety rating.

Residual risk: The demonstrator shows the functional principle only. It is explicitly not suitable for the protection of persons.

5 Threat identification

5.1 Identified threats and countermeasures

1. Threat 1:
2. Threat 2:
3. Threat 3:

5.2 Identified threats without countermeasures

1. Threat 1:
2. Threat 2:
3. Threat 3:

6 Requirements

6.1 Safety related requirements

1. Requirement:

At program start all safety related functions must be tested.

Folge Requirements: ?? , ??

1.1. Requirement

At program start the LED indicating an error must be tested.??

1.2. Requirement

At program start the USART sending error messages must be tested.

2. Requirement:

3. Requirement:

6.2 Security related requirements

6.3 Requirements with no influence on Safety and Security